C ore System

Presentation Layer (C ontroller)

Business Logic Layer (Services)

SigmaPay

SigmaPayFacade

accountService: IAccountService
budgetService: IBudgetService
goalService: IGoalService
groupSavingsService: IGroupSavingsService
paymentsService: IPaymentsServiceBL
reportingService: IReportingService
notificationService: INotificationServiceBL

handle(request): Response
createAccount(data): Response
login(credentials): Response
createBudget(plan): Response
createGoal(goal): Response
processPayment(request): Response
generateReport(filter): Report
sendNotification(message): void

I9output I8input

output input

Controller

accountService: AccountService
budgetService: BudgetService
goalService: GoalService
groupSavingsService: GroupSavingsService
paymentsService: PaymentsServiceBL
reportingService: ReportingService
notificationService: NotificationServiceBL

IControllerAuth

boolean ctlCreateAccount(String username, String email, String password, String phoneNumber)
boolean ctlUpdateUserInfo(String userId, String newEmail, String newPhoneNumber, String newAddress)
boolean ctlDeleteAccount(String userId)
boolean ctlChangeUsername(String userId, String newUsername)
boolean ctlResetPassword(String userId, String oldPassword, String newPassword)
boolean ctlVerifyIdentity(String userId, String verificationCode, String method)
UserAccount ctlGetAccountDetails(String userId)
boolean ctlEnableAccount(String userId)
boolean ctlDisableAccount(String userId)

IControllerBudget

BudgetPlan ctlCreateBudget(String userId, double totalAmount, String startDate, String endDate, List<String> categories)
boolean ctlRecordExpense(String userId, double amount, String category, String date, String paymentMethod)
boolean ctlSendBudgetAlert(String userId, String alertType, String message)

IControllerGoals

FinancialGoal ctlSetGoal(String userId, String goalType, double targetAmount, String deadline, String priorityLevel)
GoalProgress ctlTrackProgress(String userId, String goalId, boolean detailed)
AIRecommendation ctlGetOptimizedGoalPlan(String userId, String financialData, int timeFrame, boolean riskSensitive)

IControllerGroupSavings

SavingsGroup ctlCreateGroup(String groupName, List<String> members)
boolean ctlAddMember(String groupId, String memberId)
boolean ctlProcessDeposit(String groupId, String memberId, double amount)
String ctlGenerateGroupReport(String groupId)

IControllerReports

String ctlGenerateIncomeStatement(String userId)
String ctlGenerateMonthlySummary(String userId)
boolean ctlExportToPDF(String reportType)

IControllerPayments

boolean ctlProcessPayment(String userId, String methodId, double amount, String merchantId)
boolean ctlRequestRefund(String userId, String transactionId, double amount, String reason)
boolean ctlVerifyPaymentMethod(String userId, String methodId, String CVV)

IControllerNotifications

boolean ctlSendAlert(string userId, string message)
boolean ctlSetNotificationPreference(string userId, string preferenceType, boolean isEnabled)
List<Notification> ctlGetNotificationHistory(string userId)

IControllerInput

void handleRequest(string userId, string requestType, string requestData)
void handleAction(string userId, string actionType)

IControllerOutput

void respond(string userId, string responseData)
void fail(string userId, string errorMessage)

IAccountService

String register(String username, String email, String phone, String rawPassword)
boolean verifyIdentity(String userId, String code, String method)
boolean updateProfile(String userId, String email, String phone, String address)
boolean changePassword(String userId, String oldPass, String newPass)
boolean resetPassword(String userId, String newPass)
boolean enable2FA(String userId, String method)
List<ILoginActivity> getLoginActivity(String userId)

AccountService

IBudgetService

String createBudget(String userId, double total, Date start, Date end, List<String> categories)
String recordExpense(String userId, double amount, String category, Date at, String method)
boolean sendAlertIfExceeded(String userId, String period)
ISpendingTrends getTrends(String userId, String period)

BudgetService

IGoalService

String setGoal(String userId, String type, double target, Date deadline, String priority)
IGoalProgress trackProgress(String userId, String goalId, boolean detailed)
boolean adjustPlan(String goalId, double newTarget, Date newDeadline)

GoalService

IGroupSavingsService

String createGroup(String name, List<String> members)
boolean addMember(String groupId, String memberId)
String contribute(String groupId, String memberId, double amount, Date at)
IGroupReport generateReport(String groupId)
boolean requestWithdrawal(String groupId, String memberId, double amount)

GroupSavingsService

IReportingService

String generateIncomeStatement(String userId, String period)
String generateMonthlySummary(String userId, String month)
boolean exportToPDF(String reportId)
IComparisonReport comparePeriods(String userId, String p1, String p2)

ReportingService

IPaymentsServiceBL

String addBill(String userId, String provider, String accountRef, IMoney amount, Date dueAt)
boolean schedulePayment(String userId, String billId, Date when, boolean recurring)
boolean processPayment(String userId, String methodId, IMoney amount, String merchantId)
boolean requestRefund(String userId, String txnId, IMoney amount, String reason)
boolean verifyPaymentMethod(String userId, String methodId, String cvv)

PaymentsServiceBL

INotificationsServiceBL

boolean sendAlert(String userId, String message, String channel)
boolean setPreference(String userId, String type, boolean enabled)
List<INotification> history(String userId)
boolean scheduleBillReminder(String userId, String billId, Date at)

NotificationsServiceBL